

**SIMATS ENGINEERING
ASSESSMENT TOOL 3
CO 4**

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- 1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.**

Star Topology for a 20-Person Startup :

A star topology is ideal for a 20-person startup because all devices are connected to a central switch, making the network easy to manage and troubleshoot. A 24-port managed Layer 2/Layer 3 switch can connect workstations, printers, and VoIP phones while leaving extra ports for future expansion.

Each device can be connected using Cat6 UTP cables, which support Gigabit Ethernet over distances of up to 100 meters. The central switch connects to a router/firewall, which provides Internet access, DHCP, and NAT. A Wi-Fi access point connected to the switch can provide wireless connectivity for laptops and mobile devices.

The main advantage is fault isolation. If one workstation or its cable fails, only that device is affected, while the other 19 employees can continue working. Centralized connections also make maintenance, troubleshooting, and future network expansion easier.

- 2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.**

Bank Network – Recommended Topology :

A bank requires near-zero downtime, so a full or partial mesh topology is the best choice. In a full mesh, network devices have multiple direct connections, providing alternative paths if a link or device fails. In practice, banks commonly use a partial mesh at the core layer, with redundant connections between core switches, firewalls, and routers.

Dual enterprise routers provide redundant Internet connectivity using protocols such as BGP, while OSPF can automatically reroute internal traffic when a link fails. Fiber-optic cables are preferred because they provide high bandwidth, low latency, and immunity to electromagnetic interference. Single-mode fiber can be used between buildings, while multi-mode OM4 is suitable for data centers.

Additional UPS systems, redundant power supplies, and RAID storage improve availability. Therefore, mesh topology is suitable for banking because it provides high reliability, fault tolerance, automatic failover, and continuous service, which are essential for financial transactions and data integrity

- 3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single**

common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

Classroom Lab – Recommended Topology :

For a small classroom lab with a limited budget, bus topology is a suitable choice because it requires less hardware and cabling. All computers share a single backbone cable, so there is no need for an expensive central networking device.

In a traditional bus network, RG-58 coaxial cable, BNC connectors, and 50-ohm terminators are used at both ends of the backbone to prevent signal reflection. For a modern low-cost setup, an inexpensive 8- or 16-port unmanaged switch with Cat5e/Cat6 cables can be used, although this is technically a star topology.

The main disadvantages of bus topology are collisions and the possibility of the entire network failing if the backbone cable breaks. However, for a small classroom with light activities such as internet browsing and file sharing, its low cost makes it an acceptable choice. A dedicated server is also unnecessary because the teacher's computer can provide basic file-sharing services.

- 4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or**

tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

University Campus – Recommended Topology :

For a large university campus with thousands of users across multiple buildings, a hybrid topology is the most suitable choice. It combines different topologies to provide reliability, scalability, and cost-effectiveness.

At the campus backbone, a fiber-based ring or partial mesh connects core routers and multilayer switches. High-speed single-mode fiber (10 Gbps or 40 Gbps) provides fast communication and redundancy. Each building can have dual fiber uplinks to the campus core, so a single link failure does not disconnect the building.

Inside each building, a star topology connects floor-level access switches to a distribution switch, with individual computers connected using Cat6 cables. Smaller labs can use simpler bus or tree structures where appropriate. Wireless access points can connect through PoE (Power over Ethernet), reducing additional power cabling.

The core–distribution–access three-layer architecture makes the network easy to manage and expand. New buildings can be added by connecting a new distribution switch to the campus core without significantly affecting existing networks. Therefore, hybrid topology is ideal for universities because it provides redundancy, easy management, flexibility, and campus-wide scalability.

5. **Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.**

Company with Multiple Departments Recommended Network Design :

For a company with multiple departments such as HR, Finance, Sales, and IT, a hierarchical network using switches, routers, and VLANs is recommended.

Each department is connected to a Layer 2 access switch, allowing computers and printers within the department to communicate. These switches connect to a Layer 3 switch or router, which performs inter-VLAN routing and enables communication between departments.

Separate VLANs are created for each department to improve security and reduce unnecessary broadcast traffic. A router with firewall functionality connects the internal network to the Internet and protects the company from external threats.

Redundant links and backup paths can be added to improve reliability. Proper IP addressing and subnetting make the network easier to manage and expand.

Therefore, this hierarchical design provides better performance, security, reliability, easy management, and scalability as the company grows.